

JEE MAIN REVISION MASTER: ELECTROSTATICS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 270 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Coulomb's Law, Electric Field & Dipole	217 PYQs	Forces & Fields
Very High	Gauss's Law & Electrostatic Potential	78 PYQs	Flux & Potential

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Coulomb's Law, Electric Field & Dipole	$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2}$, $E_{\text{axial}} = \frac{2kp}{r^3}$, $E_{\text{equatorial}} = \frac{kp}{r^3}$, $\vec{\tau} = \vec{p} \times \vec{E}$, $U = -\vec{p} \cdot \vec{E}$
Gauss's Law & Electrostatic Potential	$\Phi_E = \oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{in}}}{\epsilon_0}$, $E = \frac{\sigma}{2\epsilon_0}$ (sheet), $E = \frac{\lambda}{2\pi\epsilon_0 r}$ (wire), $\vec{E} = -\nabla V$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Coulomb's Law, Electric Field & Dipole (Very High Yield — 217 PYQs)

Core Principle: Electrostatic forces, electric field intensity due to point charges and dipoles, and dipole torque/energy in uniform fields.

■ **Caution / JEE Pitfall:** $E_{\text{axial}} = 2E_{\text{equatorial}}$ for a short dipole. $\vec{E}_{\text{equatorial}}$ points opposite to dipole moment $\vec{p}$.

■ **Speed Trick:** Work done in rotating dipole from θ_1 to θ_2 : $W = pE(\cos\theta_1 - \cos\theta_2)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Two point charges $q_1 = 3\mu\text{C}$ and $q_2 = -4\mu\text{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ res..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "Two short electric dipoles A and B having dipole moment p_1 and p_2 respectively are placed with their axis mutually perpendicular a..." [Ans: B]
- [2026 Morning Shift] Q#1 (MCQ): "There are three co-centric conducting spherical shells A, B and C of radii a, b and c respectively ($c > b > a$) and they are charged wi..." [Ans: D]

3.2 Gauss's Law & Electrostatic Potential (Very High Yield — 78 PYQs)

Core Principle: Gauss's flux theorem, symmetry applications (infinite sheet, cylinder, sphere), and electric potential gradient $\text{vec}(\mathbf{E}) = -\text{vec}(\nabla)V$.

■ **Caution / JEE Pitfall:** Inside a charged hollow conducting sphere, electric field is ZERO, but potential is NON-ZERO and constant: $V_{\text{(inside)}} = (kq / R)$.

■ **Speed Trick:** Electric field lines are always perpendicular to equipotential surfaces and point in the direction of steepest potential decrease.

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In electrostatics, a co..." **[Ans: B]**
- **[2026 Morning Shift] Q#1 (MCQ):** "Two point charges $2q$ and q are placed at vertex A and centre of face CDEF of the cube as shown in figure. The electric flux passing ..." **[Ans: B]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Two metal plates (A, B) are kept horizontally with separation of $((12 / \pi))$ cm , with plate A on the top. An atomizer jet sprays oil..." **[Ans: A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

Two point charges $q_1 = 3\mu\text{C}$ and $q_2 = -4\mu\text{C}$ are placed at points $(2\hat{i} + 3\hat{j} + 3\hat{k})$ and $(\hat{i} + \hat{j} + \hat{k})$ respectively. Force on charge q_2 is ____ N. (Take $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ SI Units)

- (A) $(12\hat{i} + 24\hat{j} + 24\hat{k}) \times 10^{-3}$
- (B) $(4\hat{i} + 8\hat{j} + 8\hat{k}) \times 10^{-3}$ **[Correct]**
- (C) $(3\hat{i} + 6\hat{j} + 6\hat{k}) \times 10^{-3}$
- (D) $(-4\hat{i} - 8\hat{j} - 8\hat{k}) \times 10^{-3}$

Correct Answer: B

Solution Analysis: Coulomb's Law states that the force between two point charges is proportional to the product of the charges and inversely proportional to the square of the distance between them. In vector form, the force $\text{vec}(\mathbf{F})_{(21)}$ exerted on charge q_2 by charge q_1 is: $\text{vec}(\mathbf{F})_{(21)} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{|\text{vec}(\mathbf{r})|^2} \hat{r}$ Where, $\text{vec}(\mathbf{r}) = \text{vec}(\mathbf{r})_{(2)} - \text{vec}(\mathbf{r})_{(1)}$ is the position vector from q_1 to q_2 , $\hat{r} = \frac{\text{vec}(\mathbf{r})}{|\text{vec}(\mathbf{r})|}$ is the unit vector and $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ Substituting the unit vector, we get: $|\text{vec}(\mathbf{F})_{(21)}| = \frac{1}{\dots}$

Benchmark Problem 2: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R Assertion A : In electrostatics, a conductor does not store any net charge inside. Reason R : Inside the capacitor (with no dielectric medium), the free charge carriers, if placed between the plates of capacitor, experience force and drift. Choose the correct answer from the options given below

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is NOT the correct explanation of A **[Correct]**
- (C) A is true but R is false
- (D) A is false but R is true

Correct Answer: B

Solution Analysis: In electrostatic equilibrium, the electric field inside a conductor is zero ($E=0$). According to Gauss's Law ($\oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{\text{in}}}{\epsilon_0}$), if the electric field is zero everywhere inside, the net charge enclosed (q_{in}) by any Gaussian surface inside the conductor must also be zero. Therefore, any excess charge resides entirely on the outer surface. Therefore, assertion A is true. Inside a charged capacitor, there is a uniform electric field between the plates. If free charge carriers (like electrons or ions) are placed in this region, they will experience an elec